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# Quantum neural operators with implicit quadratic frame and expressivity advantages

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While classical neural operators have transformed the solving of partial differential equations, developing efficient quantum counterparts remains challenging due to processing overheads and theoretical gaps. Specifically, existing quantum machine learning paradigms often demand prohibitive qubit scaling or deep circuits, rendering them impractical for near-term hardware. Here, to bridge this gap, we propose QuanONet, a quantum neural operator tailored for the noisy intermediate-scale quantum era. Theoretically, we extend the universal approximation theorem to the quantum domain for continuous nonlinear operators. Departing from conventional views relying on exponential Hilbert spaces, we prove that the architecture's density matrix implicitly constructs a quadratic feature frame. We establish that, for operators, this implicit frame yields an $\mathcal{O}(p^2)$ expressivity bound, circumventing the $\mathcal{O}(p)$ linear capacity limits of matched classical models. To overcome the depth cost of spectrum alignment, we introduce a trainable-frequency (TF-QuanONet) strategy that adaptively spaces base frequencies to capture relatively high frequencies without parameter inflation. Extensive benchmark experiments demonstrate that TF-QuanONet notably outperforms quantum baselines and achieves competitive accuracy against classical frameworks under strictly matched-parameter conditions. Furthermore, in high-dimensional scaling regimes ($p \to 256$), the architecture exhibits superior optimization robustness, consistently converging to the intrinsic error floor where classical baselines suffer from high variance. The physical deployment on IBM quantum processors validates its functional resilience on near-term hardware as a qualitative proof of concept.

Quantum computing is emerging as a potentially disruptive technology for solving partial differential equations (PDEs)[1–3]. Prominent advancements in this domain include the Schrödingerization technique[4,5] and various variational quantum algorithms[6,7]. Parallel to this, classical machine learning—specifically neural operators such as the deep operator network (DeepONet)[8,9] in Fig. 1a and the Fourier neural operator (FNO)[10]—has transformed PDE solving by learning resolution-robust mappings between infinite-dimensional function spaces[11,12]. These classical methods typically construct finite-dimensional latent representations based on an explicit linear frame, limiting their effective representational capacity to scale linearly with the architectural width $\mathcal{O}(p)$. Consequently, in complex physical scenarios requiring high-dimensional representations, these classical-learning-based methods often suffer from optimization instability and performance saturation due to their inherent linear frame limitations.

Motivated by the complementary strengths of these domains, exploring the intersection of quantum machine learning (QML) and neural operators presents a promising avenue for demonstrating practical quantum utility in the noisy intermediate-scale quantum (NISQ) era[13,14]. However, extending QML to operator learning remains

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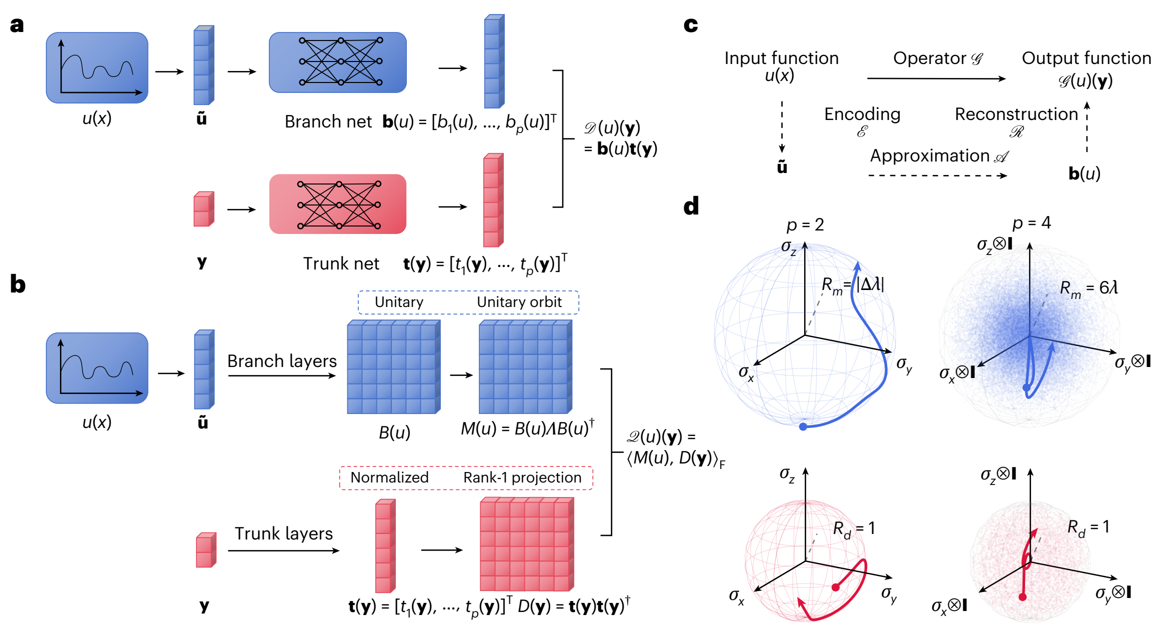


**Fig. 1 | Architectures and theoretical frameworks for neural operator learning. a**, Architecture of DeepONet, consisting of a branch net for encoding input functions and a trunk net for coordinate encoding. **b**, Architecture of QuanONet. The final operator value is computed via the Frobenius inner product. The constraints in the dashed boxes arise from quantum-mechanical properties. **c**, Schematic illustration of operator learning via the encoder–approximator–reconstructor framework. **d**, Visualization of the manifolds $D$ and $M$ projected onto the first-qubit Pauli basis. For $p=2$, the manifold is the Bloch sphere; for $p=4$, it expands into a solid volumetric ball. The spectrum is set to $\Lambda = \mathrm{diag}[-2\lambda, -\lambda, \lambda, 2\lambda]$, and unitaries are sampled from the uniform Haar measure. In both regimes, a sufficiently large spectral radius is required to cover the target codomain.

a nascent yet intellectually challenging frontier. Recent pioneering architectures, such as QDeepONet[15] and QFNO[16], face severe hardware and theoretical bottlenecks. QDeepONet employs a hybrid paradigm that restricts quantum processing strictly to linear encoding via unary states, requiring a prohibitively expensive qubit scaling of $\mathcal{O}(p)$ while lacking theoretical guarantees on representational expressivity. Conversely, QFNO requires deep circuits and $p+m$ qubits, rendering it susceptible to noise and impractical for near-term hardware. Furthermore, a critical limitation of the prevailing QML paradigm lies in treating the parameterized quantum circuit merely as a statistical generator for scalar expectation values (that is, $f(x) = \langle\psi(x)|H|\psi(x)\rangle$)[17]. This reductionist perspective overlooks the algebraic richness of quantum evolution and fails to exploit unitary operators as direct mathematical objects.

To address these critical gaps, we present QuanONet, a quantum neural operator framework tailored for the NISQ era. QuanONet effectively mitigates classical–quantum communication bottlenecks during inference and features a highly hardware-efficient logarithmic qubit scaling $\mathcal{O}(\log p)$ through amplitude embedding. Theoretically, we shift the analytical perspective from scalar function approximations to structural operator representations. By formulating the continuous nonlinear mapping via the physical measurement of expectation values—mathematically represented as the Frobenius inner product of a branch-modulated observable against a trunk-density matrix as shown in Fig. 1b—we establish two fundamental expressivity bounds. First, in the general unconstrained case, we prove that QuanONet guarantees a baseline $\mathcal{O}(p)$ approximation capacity, consistent with the linear truncation limits of classical models parameterized by a latent dimension $p$. Second, and most crucially, the density matrix inherently constructs an implicit quadratic feature frame. We establish that, for an operator whose output covariance operator has a structurally cohesive spectrum, this implicit frame successfully yields an accelerated $\mathcal{O}(p^2)$ expressivity bound, fundamentally circumventing classical capacity limits. Importantly, this native compact parameterization acts as a strong geometric regularizer, yielding superior optimization stability in high-dimensional configurations compared with the variance observed in overparameterized classical baselines.

To physically realize this theoretical $\mathcal{O}(p^2)$ advantage, the model's generated frequency difference set must perfectly align with the spectrum of the output covariance operator. In standard fixed-frequency (FF) embeddings, expanding the frequency coverage typically requires increasing circuit depth via data re-uploading, imposing a strict linear growth constraint that is prohibitive on noisy near-term hardware. To overcome this depth cost of spectrum alignment, we introduce the trainable-frequency (TF-QuanONet) strategy[18]. By treating base frequencies as continuous variables, this strategy adaptively aligns the model's spectral difference set with the output dominant modes. This effectively maximizes frequency coverage and resolves broadband dynamics without parameter inflation, bypassing the depth-induced limitations of fixed grids.

A previous exploratory work of this paper has been published as a conference paper[19]. This Article substantially extends that previous work by introducing a more profound theoretical framework and analysis; moreover, the empirical evaluation has been entirely redesigned, featuring markedly richer benchmarks alongside comprehensive comparisons across varying parameter capacities and qubit scales.

In summary, our main contributions are threefold: (1) the formulation of QuanONet and the rigorous establishment of its expressivity bounds, demonstrating that its implicit quadratic frame can elevate to an $\mathcal{O}(p^2)$ advantage; (2) the introduction of the TF strategy to adaptively align spectral difference sets, preventing dimensional collapse while strictly circumventing the circuit-depth overheads of FF grids; (3) extensive empirical validation of the architecture's representational superiority and optimization robustness, culminating in qualitative proofs of concept on physical superconducting quantum processors.

## Results

We establish the theoretical foundation of QuanONet by extending the universal operator approximation theorem to the quantum domain. Classical operator theories use finite dot products in Euclidean space;

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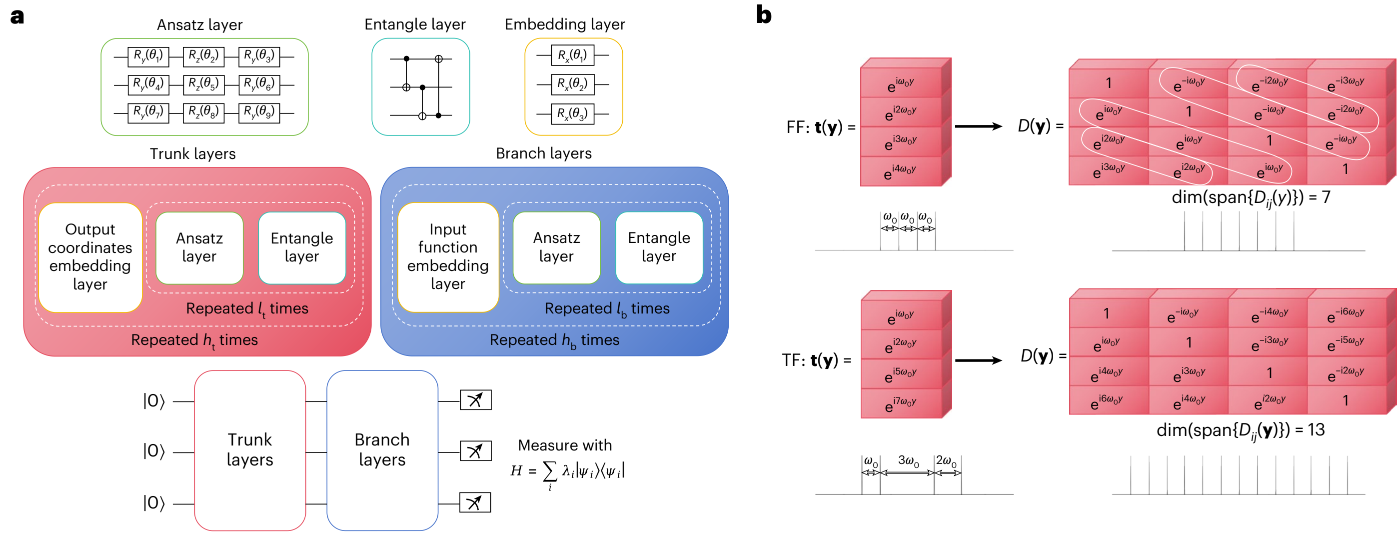


**Fig. 2 | Detailed architecture of QuanONet and comparison of trunk frequency distributions. a**, Detailed architecture of QuanONet. **b**, Illustrative visualization of the trunk-layer output state for a latent dimension $p = 4$. State normalization is omitted to highlight the spectral distribution. The FF model employs harmonic frequencies; the resulting density matrix $D(\mathbf{y})$ displays high redundancy (indicated by white ellipses) due to frequency overlaps. In contrast, the TF model yields a dense frame of dimension 13 through Wichmann construction.

these inner-product structures embed geometrically into quantum Hilbert space. However, realizing this in a physically efficient quantum protocol is challenging. Direct swap-test implementations demand duplicate registers and ancillas, causing prohibitive overhead. Moreover, projective measurements yield probabilities with the form of squared state overlaps, misaligned with the linear summation in classical approximation theorems.

To overcome these constraints without deep quantum subroutines, we formulate a theory based on the trace form of expectation values. We show that the Frobenius inner product of a modulated observable and a density matrix recovers the required linear structure with superior hardware efficiency. This formalism guarantees that a quantum system of dimension $p$ can universally approximate continuous nonlinear operators.

## Model overview

To formalize our architecture, we first establish the universal approximation property via quantum trace measurements.

**Theorem 1.** Universal approximation via quantum measurement. *Let $K_1$ be a compact subset of a Banach space of input functions, and $D_2 \subset \mathbb{R}^{d_2}$ be a compact domain of output coordinates. Let $\mathcal{G} : K_1 \to C(D_2)$ be a continuous nonlinear operator. For any $\epsilon > 0$, there exist*

1. *an observable $M(u)$ that encodes the input function $u$ with the fixed spectrum $\Lambda$ and*
2. *a rank-1 density matrix $D(\mathbf{y})$ that encodes the output coordinate $\mathbf{y}$, derived from a pure quantum state $|t(\mathbf{y})\rangle \in \mathcal{H}$,*

*such that the operator $\mathcal{G}$ can be approximated by their Frobenius inner product*:

$$\sup_{u \in K_1, \mathbf{y} \in D_2} |\mathcal{G}(u)(\mathbf{y}) - \langle M(u), D(\mathbf{y})\rangle_F| < \epsilon. \tag{1}$$

Driven by this theoretical paradigm, we propose QuanONet, a native quantum neural operator framework. By encoding the output coordinates $\mathbf{y}$ and the discretized input data $\bar{\mathbf{u}}$ into the state preparation and measurement bases of a single quantum register, respectively, the architecture evaluates the target operator using only $\mathcal{O}(\log p)$ qubits. This exponential compression of the latent space greatly reduces the circuit width requirement, preserving theoretical approximation guarantees while ensuring feasibility on near-term quantum hardware.

As shown in Fig. 2a, QuanONet comprises two primary parameterized modules: the branch and trunk layers. Rather than mapping inputs to classical latent vectors, these quantum layers generate structured operators within the latent Hilbert space $\mathcal{H} \simeq \mathbb{C}^p$. Specifically, the trunk layers encode the output coordinates $\mathbf{y}$ into a pure quantum state $|\mathbf{t}(\mathbf{y})\rangle$. This state mathematically defines a rank-1 projection matrix, which we term the trunk-density matrix:

$$D(\mathbf{y}) = \mathbf{t}(\mathbf{y})\mathbf{t}^{\dagger}(\mathbf{y}). \tag{2}$$

Conversely, the branch layers process the input function $u$ to generate a parameterized unitary transformation $B(u) \in \mathrm{SU}(p)$. Applied to a base measurement Hamiltonian characterized by a fixed diagonal spectrum $\Lambda = \mathrm{diag}[\lambda_1, \ldots, \lambda_p]$, this unitary transformation produces the branch-modulated observable via spectral conjugation:

$$M(u) = B(u)\ \Lambda B^{\dagger}(u). \tag{3}$$

The final operator approximation is then formulated as the expectation value of this modulated observable with respect to the trunk quantum state. Mathematically, this is expressed as the Frobenius inner product of the two generated matrices, supplemented by a trainable scalar bias $b_0$:

$$\mathcal{Q}(u)(\mathbf{y}) = \langle M(u), D(\mathbf{y})\rangle_F + b_0 = \mathrm{Tr}[M(u)D(\mathbf{y})] + b_0 \approx \mathcal{G}(u)(\mathbf{y}). \tag{4}$$

This formulation aligns abstract operator approximation theory directly with native quantum measurement protocols. Crucially, the expressivity advantage of QuanONet is primarily manifested within the reconstruction phase in Fig. 1c. Beyond merely mimicking classical linear combinations, the trace contraction $\mathrm{Tr}[M(u)D(\mathbf{y})]$ functions as an implicit quadratic reconstructor that naturally computes pairwise products between the latent state amplitudes. This underlying mechanism constructs a quadratic extension of the feature space during reconstruction.

As a general framework, QuanONet admits various hardware-specific implementations. To navigate the stringent constraints of the

NISQ era—such as restricted qubit connectivity and limited coherence times—we employ a hardware-efficient ansatz (HEA). This circuit design utilizes alternating layers of trainable rotations and fixed entangling gates, striking a vital balance between maximizing the reachable unitary orbit for expressivity and minimizing gate overhead to ensure practical noise resilience.

### Implicit quadratic frame and expressivity advantages

We first analyse the underlying geometric constraints. The density matrix $D(\mathbf{y})$ is confined to the complex projective space $\mathbb{C}P^{p-1}$, while the observable $M(u)$ is restricted to the unitary orbit $\mathcal{M}$ generated by the Hamiltonian's spectrum. Maximizing expressivity requires a non-degenerate spectrum, which expands the orbit dimension toward its theoretical upper bound of $p^2-p$. Visualizations in a two-qubit ($p=4$) regime demonstrate that the projected manifold of the quantum observable fills a solid volumetric region in the Pauli basis, as shown in Fig. 1d. This indicates that, despite the rigid norm constraints of quantum mechanics, the model leverages the spectral radius $\rho(\Lambda)$ to scale its representational capacity, effectively encompassing the target codomain.

Crucially, this geometric volumetric expansion in the latent space directly translates into an accelerated spectral coverage in the functional domain. Because the final output is computed via the trace operation over the quadratic density matrix, the architecture naturally synthesizes pairwise combinations of the base frequencies. We can explicitly formalize the reconstructor feature spaces for both classical and quantum architectures. For a classical DeepONet utilizing a set of trunk basis functions $\{t_1, ..., t_p\}$, the explicit linear feature space is defined as

$$V_{\mathscr{D}} = \operatorname{span}\{t_1, \ldots, t_p\} \subset L^2(D_2), \qquad \dim V_{\mathscr{D}} \leq p. \tag{5}$$

In stark contrast, the quantum trace measurement expands the basis through complex conjugations and multiplications, yielding the implicit quadratic space:

$$V_{\mathscr{Q}} = \operatorname{span}\{\Re(t_i t_j^*), \Im(t_i t_j^*) : 1 \leq i,j \leq p\} \subset L^2(D_2), \qquad \dim V_{\mathscr{Q}} \leq p^2. \tag{6}$$

This structural mechanism allows us to rigorously bound the reconstruction error, revealing a stark contrast between classical and quantum capacities.

**Theorem 2.** Generalized reconstruction error bounds. *Let $\Gamma_{\mathcal{G}_{\#}\mu}$ be the output covariance operator* of a *target functional mapping, with sorted eigenvalues $\{\lambda_m\}_{m=1}^{\infty}$ and a corresponding frequency support $\Xi$. Given a latent dimension p, the minimum mean-square reconstruction error for a standard DeepONet is strictly lower-bounded by the linear spectral tail*:

$$\min \hat{\mathcal{E}}_{\mathscr{R}}(V_{\mathscr{D}}) \geq \sum_{m=p+1}^{\infty} \lambda_m, \tag{7}$$

*as its explicit linear frame $V_{\mathscr{D}}$ spans at most p independent basis modes. For QuanONet, assuming the dominant target spectrum $\Xi_{p^2}$ (corresponding to the top $p^2$ eigenvalues) possesses a bounded additive structure with a small doubling constant ($|\Xi_{p^2}+\Xi_{p^2}| \leq K|\Xi_{p^2}|$), the implicit quadratic frame $V_{\mathscr{Q}}$ captures $\lfloor \kappa p^2 \rfloor$ independent modes from $\Xi_{p^2}$, where $\kappa \in (0,1]$ has a positive lower bound depending only on K. Consequently, its reconstruction error is rigorously bounded by*

$$\sum_{m=\lfloor \kappa p^2 \rfloor+1}^{\infty} \lambda_m \leq \min \hat{\mathcal{E}}_{\mathscr{R}}(V_{\mathscr{Q}}) \leq \sum_{m=1}^{p^2-\lfloor \kappa p^2 \rfloor} \lambda_m + \sum_{m=p^2+1}^{\infty} \lambda_m \tag{8}$$

*which represents a substantial theoretical reduction compared with the classical baseline.*

The theoretical bounds established above illuminate the core advantage of our architecture from a fundamental combinatorial perspective. In summary, substituting explicit linear features with implicit quadratic frames mitigates the representational bottleneck inherent in classical operator networks. By translating the quantum trace measurement into a combinatorial difference set in the frequency domain, QuanONet rigorously overcomes linear capacity limits. Provided with a property typical of physical systems governed by standard PDEs, which naturally form dense low-frequency clusters, this $\mathcal{O}(p^2)$ scaling enables the reconstruction of more effective components.

### TF embedding strategy

While the quadratic density matrix provides the theoretical capacity for $\mathcal{O}(p^2)$ expressivity, its practical realization depends strictly on the configuration of the base frequency parameters $\Omega$. This configuration fundamentally dictates the effective spectral coverage of the network.

Standard FF embeddings typically restrict these parameters to a predefined, uniformly spaced integer grid (for example, $\Omega = \{\omega_0, 2\omega_0, ...\}$). Under this rigid arrangement, the generated frequency difference set ($\Omega - \Omega$) exhibits severe structural redundancy. Because multiple frequency pairs yield identical differences, the maximum reachable frequency span scales strictly linearly with the re-upload times. Capturing broadband or high-frequency dynamics under an FF scheme thus necessitates an excessively large depth to brute-force the required spectral range, causing severe overheads.

To circumvent this representational bottleneck, we propose the TF-QuanONet strategy. By parameterizing $\Omega \subset \mathbb{R}$ as continuous, learnable variables, the architecture treats the frequency arrangement as a dynamic optimization problem. This continuous relaxation allows the network to adaptively space its base frequencies to directly minimize degeneracy and expand the combinatorial footprint of the difference set (Fig. 2b).

As rigorously established in Theorem 2, optimizing the spacing of $p$ parameters enables their combinatorial differences to effectively encompass $\mathcal{O}(p^2)$ distinct spectral modes. By aligning its adaptive quadratic frame directly with the structurally cohesive eigenfrequencies of the output covariance operator, the TF-QuanONet strategy ensures the precise resolution of dominant high-frequency bands without artificially inflating the network size.

### Experimental benchmarks and baselines

To evaluate the empirical performance of our framework, we benchmark TF-QuanONet across three ordinary differential equation (ODE) operators—comprising the antiderivative, homogeneous and nonlinear operators—and three PDE solution operators that govern distinct physical dynamics. These PDE systems include the source-term-dominated diffusion–reaction (D–R) system, the linear transport of the advection equation and the steady-state potential field of the Darcy flow. The core governing relations and operator mappings $\mathcal{G} : u \mapsto v$ are formulated as follows.

$$\text{Antiderivative} : \frac{\mathrm{d}v(x)}{\mathrm{d}x} = u(x), \quad v(0) = 0, \tag{9}$$

$$\text{Homogeneous} : \frac{\mathrm{d}v(x)}{\mathrm{d}x} = v(x) + u(x), \quad v(0) = 0. \tag{10}$$

$$\text{Nonlinear} : \frac{\mathrm{d}v(x)}{\mathrm{d}x} = -v(x)^3 + u(x), \quad v(0) = 0. \tag{11}$$

$$\text{Diffusion dash reaction} : \frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2} + kv^2 + u, \quad v(x,0) = 0, D = 0.01, k = 0.01. \tag{12}$$

**Table 1 | Relative $L_2$ error comparison on benchmarks**

| Method | $\omega_0$ | ODE | | |
|---|---|---|---|---|
| | | Antiderivative operator | Homogeneous operator | Nonlinear operator |
| HEA | 0.001 | 71.789 ± 0.899% | 81.474 ± 1.054% | 69.913 ± 1.385% |
| | 0.01 | 53.540 ± 0.601% | 56.212 ± 0.479% | 53.994 ± 0.867% |
| | 0.1 | 40.233 ± 0.564% | 45.838 ± 0.777% | 38.440 ± 0.896% |
| TF-HEA | 0.001 | 11.557 ± 0.072% | 8.047 ± 0.082% | 13.250 ± 0.144% |
| | 0.01 | 11.634 ± 0.102% | 8.040 ± 0.068% | 13.292 ± 0.173% |
| | 0.1 | 11.638 ± 0.091% | 8.054 ± 0.056% | 13.262 ± 0.195% |
| QuanONet | 0.001 | 83.908 ± 1.778% | 90.596 ± 1.600% | 82.652 ± 1.048% |
| | 0.01 | 46.370 ± 1.559% | 51.806 ± 1.825% | 46.618 ± 1.435% |
| | 0.1 | 7.793 ± 0.542% | 6.627 ± 0.725% | 8.754 ± 0.287% |
| TF-QuanONet | 0.001 | 2.376 ± 0.186% | 2.335 ± 0.225% | 2.911 ± 0.298% |
| | 0.01 | 2.492 ± 0.071% | 2.396 ± 0.281% | 2.959 ± 0.137% |
| | 0.1 | 2.160 ± 0.194% | 2.196 ± 0.176% | 2.635 ± 0.257% |
| DeepONet | | 5.162 ± 1.167% | 3.538 ± 0.710% | 6.099 ± 1.134% |
| FNN | | 6.233 ± 1.606% | 4.699 ± 1.806% | 7.211 ± 2.344% |
| | | PDE | | |
| | | D–R system | Advection equation | Darcy flow |
| HEA | 0.001 | 80.109 ± 0.683% | 83.948 ± 0.842% | 72.386 ± 0.622% |
| | 0.01 | 79.150 ± 0.704% | 76.873 ± 0.577% | 62.112 ± 0.617% |
| | 0.1 | 63.457 ± 1.505% | 77.057 ± 0.697% | 43.035 ± 0.383% |
| TF-HEA | 0.001 | 12.874 ± 0.645% | 66.017 ± 11.977% | 7.950 ± 0.113% |
| | 0.01 | 12.909 ± 0.459% | 48.645 ± 17.764% | 7.925 ± 0.170% |
| | 0.1 | 12.547 ± 0.269% | 32.432 ± 0.678% | 7.731 ± 0.203% |
| QuanONet | 0.001 | 80.473 ± 0.869% | 91.023 ± 1.257% | 81.755 ± 1.053% |
| | 0.01 | 69.209 ± 0.905% | 76.956 ± 0.525% | 57.249 ± 0.542% |
| | 0.1 | 31.099 ± 0.861% | 50.424 ± 2.041% | 11.977 ± 0.058% |
| TF-QuanONet | 0.001 | 7.589 ± 0.364% | 29.374 ± 1.733% | 6.822 ± 0.185% |
| | 0.01 | 7.733 ± 0.484% | 27.028 ± 3.348% | 6.914 ± 0.232% |
| | 0.1 | 7.196 ± 0.236% | 16.322 ± 0.719% | 6.628 ± 0.181% |
| DeepONet | | 15.462 ± 0.546% | 9.994 ± 0.878% | 5.674 ± 0.408% |
| FNN | | 16.844 ± 1.521% | 29.315 ± 3.501% | 9.083 ± 0.609% |

Data are presented as mean ±s.d. derived from $n$=5 independent trials with randomized network initializations and data splits.

$$\text{Advection}: \frac{\partial v}{\partial t} = -c\frac{\partial v}{\partial x}, \quad v(x,0) = u(x), c = 1. \tag{13}$$

$$\text{Darcy flow}: -\nabla(K\nabla v) = f, \quad v|_{\partial D_2} = u, K = 0.1, f = -1. \tag{14}$$

All computational domains are normalized to the unit interval [0, 1], while comprehensive details regarding sensor configurations and data generation protocols are deferred to Methods. To establish a rigorous evaluation, the proposed framework is compared against established quantum methods, including the standard HEA and the trainable-frequency HEA (TF-HEA), alongside classical architectures (fully connected neural networks (FNN) and DeepONet). Notably, the classical DeepONet serves as a performance proxy for the recently proposed hybrid QDeepONet. Because the latter primarily utilizes quantum circuits to accelerate the forward propagation phase rather than structurally expanding the underlying representational frame, the classical DeepONet provides a reliable representational upper bound for its hybrid quantum counterpart.

### Experimental evaluation on matched-parameter benchmarks

As summarized in Table 1, empirical evaluations under strictly matched-parameter conditions (~1,200 parameters for ODEs and ~2,400 for PDEs) reveal task-dependent performance variations. On the ODE benchmarks, vanilla QuanONet exhibits pronounced sensitivity to the initial frequency scale $\omega_0$, supporting our theoretical analysis regarding the dimensional collapse of FF embeddings. In contrast, by dynamically adapting its spectral alignment, TF-QuanONet demonstrates enhanced robustness and consistently outperforms both classical DeepONet and the TF-HEA baselines. For PDE operators, TF-QuanONet yields a relative error of 7.20% on the nonlinear diffusion–reaction system, substantially outperforming classical DeepONet (15.46%) by effectively capturing source-driven reaction fields. Conversely, transport-dominated regimes—represented by the advection equation and Darcy flow—exhibit markedly more marked operator responses characterized by highly broadband spectral features. Compared with source-driven systems, resolving these sharp, localized wavefronts demands a vastly wide and meticulous spectral coverage. Consequently, under the highly restricted parameter budget,

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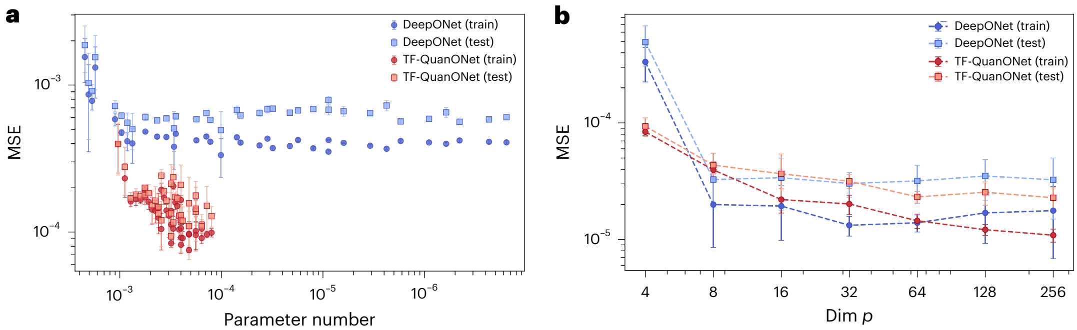


**Fig. 3 | Performance and scaling analysis of TF-QuanONet. a**, Train–test error of TF-QuanONet and DeepONet versus parameter counts. **b**, Scaling behaviour of train–test mean squared error (MSE) with respect to $p$, sweeping from $p = 4$ to $p = 256$ (corresponding to 2–8 qubits). The curves report the optimal average performance achieved across extensive network size configurations. Data are presented as mean ± s.d. derived from $n = 5$ independent trials with randomized network initializations and data splits.

the extreme compactness of the quantum architecture encounters an expressivity bottleneck. Visualizations of the predictions across distinct input frequencies are provided in Extended Data Fig. 1.

To rigorously probe the scalability of representational density, we extend our evaluation to a high-capacity regime (~10,000 parameters) to include the FNO. To accommodate the fast Fourier transform mechanics inherent to the FNO, we establish a unified framework by uniformly aligning the input data sampling to $m = 100$ sensors. The comparative error metrics detailed in Extended Data Table 1 reveal a nuanced performance landscape under this symmetric parameterization. On the linear ODE benchmarks (for example, antiderivative and homogeneous), TF-QuanONet demonstrates superior expressivity, achieving test errors of 3.12% and 3.21%, which surpass both the FNO and DeepONet. However, on the nonlinear ODE benchmark, TF-QuanONet exhibits performance degradation compared with its classical counterparts. This disparity suggests that, while the implicit quadratic frame scales robustly for linear dynamics, the current quantum architecture lacks the structural capacity to effectively capture the highly complex spectra and basis functions of nonlinear operators under aligned data sampling.

## Expressivity verification and high-dimensional scaling dynamics

To empirically validate the representational advantage of the implicit quadratic frame, we first evaluate the architectures under a strictly constrained latent dimension $p = 4$ (equivalent to two qubits) on the antiderivative operator. As detailed in Fig. 3a, the classical DeepONet encounters an empirical performance plateau; even when scaled to an overparameterized regime of over $6.4 \times 10^6$ parameters, its relative error saturates at approximately 5.5%, accompanied by a generalization gap indicative of overfitting. In contrast, TF-QuanONet overcomes this linear representation bottleneck, achieving relative errors of approximately 2.5% with a negligible generalization gap under a compact parameter budget (~$10^3$).

To further investigate the limits of operator approximation, we scale the latent dimension $p$ from 4 to 256 (corresponding to 2 to 8 qubits) across various network configurations. As visualized in Fig. 3b, the two frameworks converge to an empirical error floor of approximately $2 \times 10^{-5}$ at $p = 8$, probably bounded by the discrete encoding precision. However, a clear divergence in optimization stability emerges in higher-dimensional regimes ($p > 32$). While DeepONet exhibits variance across different random seeds—with test errors fluctuating between $2 \times 10^{-5}$ and $4 \times 10^{-5}$ due to initialization sensitivity—TF-QuanONet maintains high numerical stability. Even at $p = 256$, the quantum operator consistently converges to the aforementioned error floor with minimal variance across independent trials. Detailed numerical results for scaling behaviours across varying configurations are provided in Extended Data Tables 2 and 3. The empirical consistency suggests that the native quantum parameterization provides inherent structural regularization. This mechanism effectively mitigates the optimization variance commonly associated with overparameterized classical learning models in high-dimensional spaces.

## Circuit architecture trade-offs: width and depth

We further isolate the architectural impact of circuit complexity by analysing the trade-offs between depth ($h_b$, $h_t$) and width (qubit count) on approximation accuracy. The empirical results compiled in Extended Data Table 4 indicate that systematic increases in circuit depth consistently reduce reconstruction errors; for instance, expanding the depth from ($h_b = 20$, $h_t = 10$) to (40, 40) on a five-qubit system reduces the relative $L_2$ error from 2.41% to 1.64%. This suggests that sufficient layer repetition is critical for capturing the complex frequency spectrum of the target operator. Conversely, the impact of circuit width is governed by a balance between representational space and trainability. While increasing the qubit count expands the latent Hilbert space dimension to offer a richer feature map, it simultaneously complicates the optimization landscape by introducing trainability barriers such as the barren plateau problem. This trade-off is evident as the ten-qubit model exhibits performance degradation, with relative errors rising to 2.05–5.75% due to optimization difficulties, whereas the five-qubit configuration achieves a more favourable balance between functional expressivity and stable convergence.

## Impact of Hamiltonian design and spectral properties

The spectral structure of the measurement Hamiltonian determines the expressivity of QuanONet by governing the geometry of the unitary orbit $\mathcal{M}_\Lambda$ into which the branch layers map input information. The algebraic dimension of this orbit is given by $\dim(\mathcal{M}_\Lambda) = p^2 - \sum_{j=1}^{q} n_j^2$, where $n_j$ represents the multiplicity of each eigenvalue. Theoretically, a strictly non-degenerate spectrum maximizes this dimension to $p^2 - p$, whereas degeneracy reduces the orbit dimension and constrains representational capacity. To empirically evaluate this mechanism, we benchmarked four Hamiltonians with varying degrees of degeneracy—ranging from a highly degenerate $\Lambda_1 = \mathrm{diag}[-5, 5, 5, 5]$ (dim = 6) to a fully non-degenerate $\Lambda_4 = \mathrm{diag}[-5, -2.5, 2.5, 5]$ (dim = 12)—on the antiderivative operator using a two-qubit system ($p = 4$). As shown in Fig. 4a, model accuracy generally correlates with the orbit dimension, with the non-degenerate $\Lambda_4$ achieving the lowest error among the tested configurations. However, $\Lambda_2 = \mathrm{diag}[-5, -5, 5, 5]$ underperforms the lower-dimensional $\Lambda_1$. This observation indicates that the Hamiltonian spectrum shapes not only the raw manifold dimension but also its specific geometric structure.

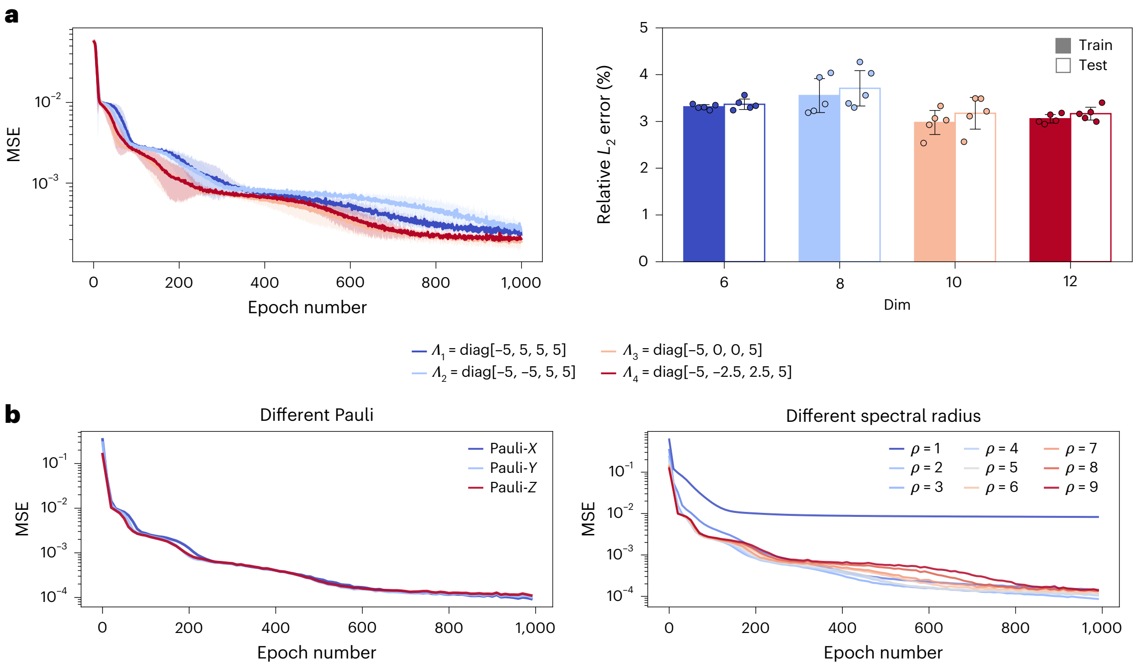


**Fig. 4 | Influence of Hamiltonian properties on model accuracy. a**, Errors of TF-QuanONet on the antiderivative operator using Hamiltonians with different spectra. The legend indicates the eigenvalues. **b**, Error curve of TF-QuanONet versus Hamiltonian Pauli terms and spectral radius for the antiderivative operator. Here, $\rho = 7$ denotes a scaled Hamiltonian $H = \frac{7}{5}\sum_{i=1}^{5}\sigma_z^{(i)}$. Data are presented as mean ± s.d. derived from $n = 5$ independent trials with randomized network initializations and data splits.

Beyond degeneracy, we investigate global spectral properties against specific basis configurations in Fig 4b. The architecture exhibits invariance to specific choices of the Pauli basis ($\sigma_x, \sigma_y, \sigma_z$), provided that the eigenspectrum is preserved. Conversely, the spectral radius $\rho(\Lambda) = \max_i |\lambda_i|$, which bounds the feasible region of achievable coefficients, acts as a critical hyperparameter. We observe a distinct threshold effect: when $\rho(\Lambda)$ is insufficient to enclose the target operator's codomain, the model is geometrically constrained and prone to underfitting. Once the spectral radius expands sufficiently to cover the target range, the accuracy plateaus, suggesting that the overall spectral magnitude is more critical than the specific basis alignment in ensuring universal expressivity.

### Real-device deployment as a proof of concept

The deployment of quantum algorithms on NISQ devices necessitates a candid evaluation of their resilience against inherent hardware noise. We investigate the physical robustness of TF-QuanONet using a superconducting quantum processor, `ibm_fez` (accessing qubits q136 and q143). To accommodate the limited coherence times, we implement a shallow circuit configuration ($h_b = h_t = 5$, $l_b = l_t = 1$) for the antiderivative operator ($m = 10$). Evaluated on linear ($u(x) = x$) and trigonometric ($u(x) = \cos \pi x$) inputs with 10,000 shots, the logical circuit (depth 60) is transpiled to the native gate set ($R_z$, $\sqrt{X}$, $CZ$) with a physical depth of 109. This manageable overhead ensures that execution largely remains within the hardware's coherence window. Detailed circuit compilation statistics for the real-device deployment are listed in Extended Data Table 5.

Quantitatively, the transition from noise-free simulation to physical hardware incurs an expected performance penalty, shifting the relative error from a ~5% baseline to the 30–40% regime. However, from a structural perspective, the hardware inference avoids degenerating into unstructured noise. As illustrated in Fig. 5, the model successfully preserves the macroscopic functional curvature and global trends of the target solutions. These results demonstrate that the hardware-efficient topology of TF-QuanONet possesses a baseline resilience, capable of capturing qualitative topological features even in the presence of unmitigated NISQ noise.

Nevertheless, the severe constraints highlighted by this numerical gap must be frankly acknowledged. In the domain of differential equation solving—where high numerical fidelity is non-negotiable—a 30–40% noise-induced error floor constitutes a critical barrier for engineering-grade applications. Consequently, while this physical deployment serves as a structural proof of concept for the architecture's expressivity, bridging the gap from macroscopic shape retention to precision scientific computing will strictly rely on the maturation of quantum error correction and fault-tolerant protocols in future hardware generations.

## Discussion

This work establishes a native quantum framework for scientific machine learning by bridging abstract operator approximation theory with hardware-efficient circuit architectures. By shifting the analytical perspective from scalar function approximation to the parameterization of continuous unitary operators on the manifold, QuanONet introduces an alternative paradigm for QML. Supported by our analysis, the trace measurement protocol against a Hermitian observable ensures physically valid real-valued outputs, while fundamentally acting as an implicit quadratic reconstructor that yields an effective representational capacity scaling as $\mathcal{O}(p^2)$.

From a theoretical standpoint, it is important to note that our rigorous $\mathcal{O}(p^2)$ expressivity bound is currently established under the assumption that the output covariance operator's optimal eigenfunctions are Fourier modes. This specific assumption enables the natural mapping of the quantum trace measurement to a combinatorial difference set in the frequency domain. While this underlying mechanism naturally extends to other standard orthogonal bases, such as Chebyshev polynomials, where the multiplication of basis functions similarly generates combinatorially expanded spans via well-defined product formulas, a fully generalized theory for completely arbitrary operators remains an open question. Establishing the precise

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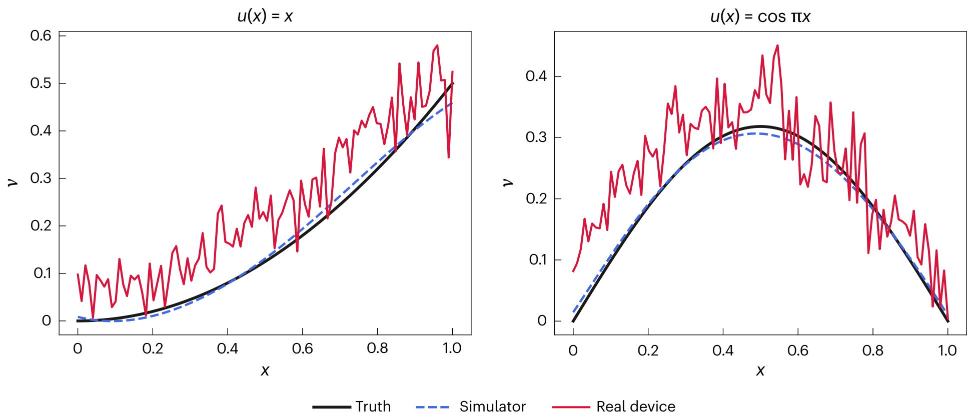


**Fig. 5 | Performance comparison between the noise-free simulator and a physical device (ibm_fez).** The figure presents a detailed inference comparison between the ideal simulations and the real hardware evaluations of TF-QuanONet for the antiderivative operator, given the linear and triangular input functions.

topological and algebraic boundaries of this quadratic advantage across arbitrary eigengeometries is a critical direction for future mathematical work.

A comprehensive evaluation requires a balanced assessment of both algorithmic capabilities and physical constraints. While TF-QuanONet demonstrates parameter efficiency on operators with globally smooth mapping profiles, such as typical diffusion–reaction processes, target mappings characterized by sharp, localized structural variations reveal certain limitations under compact parameter budgets. We hypothesize that the intense responses—typically induced by boundary and initial conditions—pose a substantial challenge for the adaptive alignment of the frequency difference set. Consequently, enhancing the frequency optimization mechanism to robustly capture these localized dynamics remains an important direction for future algorithmic refinement.

Critically, our deployment on physical superconducting hardware highlights the practical barriers imposed by NISQ-era noise. Although the physical circuit successfully captures the macroscopic functional curvature without error correction overhead, unmitigated hardware noise introduces a persistent error floor. For differential equation solving, where numerical precision is paramount, current unmitigated hardware deployments function primarily as qualitative proofs of concept. Bridging the gap from qualitative shape retention to precision scientific computing will rely heavily on the maturation of fault-tolerant quantum error correction protocols.

Looking forwards, the density-matrix perspective established here opens profound avenues for broader architectural designs. Accommodating multivariable inputs or coupled multiphysics systems could be naturally formulated by leveraging the tensor products of multiple density matrices to construct higher-order interaction frames. Transcending specific empirical benchmarks, this architecture provides a foundational theoretical and practical step towards scaling quantum neural operators for complex scientific simulations.

# Methods

## The deep operator network

Deep operator network (DeepONet)[20] is a deep learning framework designed to approximate nonlinear operators. Its architecture is theoretically grounded in the universal approximation theorem for operators[21].

DeepONet consists of two subnetworks: the branch net, which encodes the input function $u$ (discretized as $\tilde{\mathbf{u}} = [u(\mathbf{x}_1), \ldots, u(\mathbf{x}_m)]^{\mathrm{T}} \in \mathbb{R}^m$), and the trunk net, which encodes the output coordinate $\mathbf{y} \in \mathbb{R}^{d_2}$. Let $\mathcal{N}$ denote a generic deep neural network parameterized by trainable weights $\boldsymbol{\theta}$. The outputs of the branch (BNC) and trunk (TNK) nets, $\mathbf{b}(u), \mathbf{t}(\mathbf{y}) \in \mathbb{R}^p$, are defined as

$$\mathbf{b}(u) = \mathcal{N}_{\mathrm{BNC}}(\tilde{\mathbf{u}}; \boldsymbol{\theta}_{\mathrm{b}}),$$
$$\mathbf{t}(\mathbf{y}) = \mathcal{N}_{\mathrm{TNK}}(\mathbf{y}; \boldsymbol{\theta}_{\mathrm{t}}).$$

In the original formulation[8], these components are instantiated as multilayer perceptrons. The final operator approximation is obtained via the dot product:

$$\mathcal{D}(u)(\mathbf{y}) = \mathbf{b}(u) \cdot \mathbf{t}(\mathbf{y}) + b_0 \approx \mathcal{G}(u)(\mathbf{y}). \tag{15}$$

This architecture is supported by the following universal approximation theorem.

**Theorem 3.** Universal approximation of DeepONet[21]. *Let $K_1$ be a compact subset of a Banach space of input functions, $D_2 \subset \mathbb{R}^{d_2}$ be a compact domain and $\mathcal{G} : K_1 \to C(D_2)$ be a continuous nonlinear operator. For any $\epsilon > 0$, there exists a deep operator network $\mathcal{D}$ parameterized by a latent dimension $p$ such that*

$$\sup_{u \in K_1, \mathbf{y} \in D_2} \left| \mathcal{G}(u)(\mathbf{y}) - \sum_{k=1}^{p} b_k(u) t_k(\mathbf{y}) \right| < \epsilon. \tag{16}$$

DeepONet enables rapid inference on unseen initial or boundary conditions without retraining. Extensions such as the unstacked DeepONet[8], which utilizes a shared branch net to reduce computational costs, and the sequential DeepONet[22], which incorporates sequential learning for time-dependent problems, have further broadened its applicability.

## The architecture of QuanONet

On the basis of the theoretical paradigm established in 'Results', we propose QuanONet. By sequentially embedding the discretized input data $\tilde{\mathbf{u}} = [u(\mathbf{x}_1), \ldots, u(\mathbf{x}_m)]^{\mathrm{T}}$ and output coordinates $\mathbf{y}$ into a single quantum register, QuanONet evaluates the target operator using only $\mathcal{O}(\log p)$ qubits, thereby greatly reducing circuit width while preserving approximation guarantees.

The branch and trunk layers encode the input function $\tilde{\mathbf{u}}$ and the output coordinates $\mathbf{y}$, respectively. The depth of these networks is governed by the hyperparameters $h_{\mathrm{b}}$ and $h_{\mathrm{t}}$, which denote the number of repetitions of the embedding layers, while $l_{\mathrm{b}}$ and $l_{\mathrm{t}}$ represent the variational ansatz depth. The raw circuit output evaluates the expectation value of the Hamiltonian $H$:

$$\mathcal{Q}(u)(\mathbf{y}) = \langle 0 | U^{\dagger}_{\mathrm{TNK}}(\boldsymbol{\theta}^{\mathrm{t}}, \mathbf{y}) U^{\dagger}_{\mathrm{BNC}}(\boldsymbol{\theta}^{\mathrm{b}}, \tilde{\mathbf{u}}) H U_{\mathrm{BNC}}(\boldsymbol{\theta}^{\mathrm{b}}, \tilde{\mathbf{u}}) U_{\mathrm{TNK}}(\boldsymbol{\theta}^{\mathrm{t}}, \mathbf{y}) | 0 \rangle. \tag{17}$$

To bridge this physical measurement with the operator approximation framework, we express the Hamiltonian via its spectral decomposition $H = \sum_{i=1}^{p} \lambda_i |\psi_i\rangle\langle\psi_i|$. By defining the parameterized trunk state as $\mathbf{t}(\mathbf{y}) = U_{\text{TNK}}(\boldsymbol{\theta}^{\text{t}}, \mathbf{y})|0\rangle$ and the set of branch states as $|b_i(u)\rangle = U^{\dagger}_{\text{BNC}}(\boldsymbol{\theta}^{\text{b}}, \tilde{\mathbf{u}})|\psi_i\rangle$, the output fundamentally reformulates into a weighted sum of transition probabilities:

$$\mathcal{Q}(u)(\mathbf{y}) = \sum_{i=1}^{p} \lambda_i |\langle b_i(u)|t(\mathbf{y})\rangle|^2. \tag{18}$$

As formally introduced in our model overview, this expression can be elegantly abstracted into a matrix inner product. By encoding the pure trunk state into a rank-1 density matrix $D(\mathbf{y}) = \mathbf{t}(\mathbf{y})\mathbf{t}^{\dagger}(\mathbf{y})$, and assembling the branch states to modulate the Hamiltonian spectrum into the observable $M(u) = B(u)\Lambda B^{\dagger}(u)$, the quantum execution strictly reduces to a Frobenius inner product. Including a trainable classical bias $b_0$, the final operator output is efficiently evaluated as

$$\mathcal{Q}(u)(\mathbf{y}) = \langle M(u), D(\mathbf{y})\rangle_{\text{F}} + b_0 \approx \mathcal{G}(u)(\mathbf{y}). \tag{19}$$

### Hardware and software set-up

All numerical simulations were performed on a high-performance workstation equipped with an AMD Ryzen Threadripper 3970X CPU (32 physical cores), 190 GB random-access memory and five NVIDIA GeForce RTX 3090 graphics processing units. The noise-free quantum neural network simulations utilize a custom noise-free simulator built upon the MindSpore framework[23], while classical neural networks are implemented via the DeepXDE library[24].

For experimental validation on physical hardware, we utilized the IBM Quantum superconducting processor ibm_fez (accessing qubits q136 and q143). The quantum circuit transpilation, optimization and error-mitigation protocols (including measurement error mitigation) were implemented using the Qiskit SDK[25].

### Data generation

Training and testing datasets are generated from mean-zero Gaussian random fields (GRFs), $u \approx \mathcal{G}(0, k_l)$, characterized by the radial basis function covariance kernel:

$$k_l(x_1, x_2) = \exp\left(-\frac{\| x_1 - x_2 \|^2}{2l^2}\right), \tag{20}$$

where $l > 0$ is the length-scale parameter controlling smoothness. The sampled functions $u$ serve as inputs for the operator learning tasks. The target solutions are obtained using standard numerical solvers: the Runge–Kutta (4, 5) method for ODE systems and a second-order finite-difference scheme for PDEs. To construct the data pairs, for each input sample $u$, we randomly select $P$ spatial locations in the output domain to evaluate the corresponding solution. We adopt a uniform resolution of 100 across all benchmark problems, with the Darcy flow problem being the exception (resolution 25). A comprehensive summary of the dataset parameters and sampling locations is listed in Extended Data Table 6.

### Training configuration

All reported results represent the mean and s.d. computed over five independent trials with randomized data splitting and network initialization. To ensure a fair comparison, we constrain model complexity by setting the number of trainable parameters to approximately 1,200 for ODEs and 2,400 for PDEs. Optimization is performed using Adam[26] with a learning rate of $10^{-4}$. All quantum neural networks are implemented on a five-qubit system. The measurement Hamiltonian is defined as the sum of local Pauli-$Z$ operators:

$$H = \sum_{i=1}^{5} \sigma_z^{(i)}. \tag{21}$$

We adopt this Hamiltonian as the default configuration due to its hardware-efficient implementation (native virtual $Z$-gates) and its prevalence as a standard benchmark in NISQ applications. Although a non-degenerate spectrum theoretically yields higher expressivity, this degenerate set-up serves as a conservative baseline to evaluate the model's performance under restricted hardware conditions. Detailed hyperparameter configurations for the ODE, PDE and FNO baselines are provided in Extended Data Tables 7, 8 and 9, respectively. Unless otherwise specified, these settings are kept consistent across all experimental trials.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

All the datasets in the study were generated directly from the code. Source data are provided with this paper.

## Code availability

The code[27] used in the study is publicly available from the GitHub repository (https://github.com/Wang-Ruocheng/QuanONet) and has been permanently archived on Zenodo at https://doi.org/10.5281/zenodo.21084057.

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## Author contributions

R.W. conceived the model, developed the codebase, and designed and conducted the experiments. X.Z. established and proved the theoretical properties of the model. Z.X. refined the codebase and designed the deployment strategy for the real quantum hardware. J.Y. supervised the research methodology and coordinated the project. All authors reviewed and approved the final paper.

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## Competing interests

The authors declare no competing interests.

## Additional information

**Extended data** is available for this paper at https://doi.org/10.1038/s42256-026-01289-7.

**Supplementary information** The online version contains supplementary material available at https://doi.org/10.1038/s42256-026-01289-7.

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**Extended Data Table 1 | Relative $L_2$ error with FNO on ODE benchmarks under aligned data**

| Method | Antiderivative Operator | Homogeneous Operator | Nonlinear Operator |
|---|---|---|---|
| TF-QuanONet | **3.116** ± **0.492**% | **3.209** ± **0.269**% | 4.770 ± 0.201% |
| DeepONet | 6.330 ± 0.220% | 6.982 ± 1.588% | 4.040 ± 0.499% |
| FNO | 3.888 ± 0.937% | 4.057 ± 0.937% | **3.761** ± **0.621**% |

Data are presented as mean values ± SD derived from $n$ = 5 independent trials with randomized network initializations and data splits.

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**Extended Data Table 2 | Relative $L_2$ error scaling of TF-QuanONet across varying latent dimensions**

| Dim | $h_t$ | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|---|
| 4 | 50 | 4.524 ± 1.073% | 3.805 ± 0.751% | 3.018 ± 0.160% | 3.105 ± 0.169% | 3.072 ± 0.124% |
| | 100 | 3.280 ± 0.069% | 3.137 ± 0.252% | 2.842 ± 0.391% | 2.944 ± 0.401% | 2.673 ± 0.199% |
| | 150 | 3.602 ± 0.196% | 3.187 ± 0.125% | 2.959 ± 0.215% | 2.639 ± 0.441% | 2.642 ± 0.549% |
| | 200 | 3.551 ± 0.337% | 3.052 ± 0.138% | 2.981 ± 0.501% | 2.607 ± 0.421% | 2.617 ± 0.411% |
| 8 | 100 | - | 2.168 ± 0.198% | - | 2.000 ± 0.437% | 1.840 ± 0.136% |
| | 200 | - | 2.275 ± 0.462% | - | 1.789 ± 0.146% | 1.764 ± 0.138% |
| 16 | 100 | - | - | - | - | 1.477 ± 0.100% |
| | 200 | - | - | - | - | 1.498 ± 0.151% |
| 32 | 100 | - | - | - | - | 1.400 ± 0.123% |
| 64 | 100 | - | - | - | - | 1.323 ± 0.061% |
| 128 | 100 | - | - | - | - | 1.237 ± 0.129% |
| 256 | 100 | - | - | - | - | 1.169 ± 0.097% |
| **Dim** | **$h_t$** | **60** | **100** | **150** | **200** | **300** |
| 4 | 50 | 3.091 ± 0.251% | 2.800 ± 0.381% | 3.332 ± 0.690% | 3.309 ± 0.633% | 3.065 ± 0.419% |
| | 100 | 2.509 ± 0.455% | 2.235 ± 0.203% | 2.517 ± 0.452% | 2.833 ± 0.722% | 2.424 ± 0.148% |
| | 150 | 2.490 ± 0.246% | 2.478 ± 0.503% | 2.381 ± 0.546% | 2.705 ± 0.304% | 2.802 ± 0.571% |
| | 200 | 2.516 ± 0.328% | 2.407 ± 0.361% | 2.728 ± 0.623% | 2.447 ± 0.238% | 2.616 ± 0.247% |
| 8 | 100 | - | 1.663 ± 0.224% | 1.603 ± 0.126% | 1.871 ± 0.445% | 2.006 ± 0.185% |
| | 200 | - | 1.770 ± 0.352% | 1.583 ± 0.290% | 1.518 ± 0.200% | 1.497 ± 0.434% |
| 16 | 100 | - | 1.407 ± 0.088% | - | - | - |
| | 200 | - | 1.374 ± 0.317% | - | - | - |
| 32 | 100 | - | 1.301 ± 0.090% | - | - | - |
| 64 | 100 | - | 1.115 ± 0.065% | - | - | - |
| 128 | 100 | - | 1.163 ± 0.132% | - | - | - |
| 256 | 100 | - | 1.102 ± 0.122% | - | - | - |

The plot illustrates the generalization capability of the model on the Antiderivative operator benchmark. Data are presented as mean values ± SD derived from $n$ = 5 independent trials with randomized network initializations and data splits.

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**Extended Data Table 3 | Relative $L_2$ error scaling of DeepONet across varying latent dimensions**

| Dim | B&T Depth | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 4 | 4 | 9.840 ± 2.173% | 7.211 ± 2.117% | 6.982 ± 0.435% | 8.946 ± 2.015% |
| | 8 | 6.222 ± 0.311% | 5.769 ± 0.155% | 5.443 ± 0.499% | 5.137 ± 0.878% |
| | 16 | 5.704 ± 0.168% | 5.557 ± 0.108% | 5.648 ± 0.090% | 5.141 ± 1.029% |
| | 32 | 5.740 ± 0.151% | 5.608 ± 0.081% | 5.580 ± 0.134% | 5.056 ± 1.109% |
| | 64 | 5.888 ± 0.116% | 5.778 ± 0.139% | 5.892 ± 0.158% | 6.096 ± 0.127% |
| | 128 | 6.028 ± 0.212% | 5.905 ± 0.091% | 6.075 ± 0.119% | 6.521 ± 0.224% |
| | 256 | 6.066 ± 0.182% | 5.969 ± 0.252% | 5.886 ± 0.149% | 6.242 ± 0.276% |
| | 512 | 6.012 ± 0.086% | 5.512 ± 0.100% | 5.629 ± 0.108% | 5.915 ± 0.262% |
| | 1024 | 6.041 ± 0.266% | 5.499 ± 0.111% | 5.599 ± 0.092% | 5.703 ± 0.054% |
| 8 | 8 | 5.808 ± 0.155% | 4.081 ± 1.081% | 2.934 ± 0.432% | 3.240 ± 0.187% |
| | 16 | 4.001 ± 1.548% | 2.527 ± 0.643% | 2.492 ± 0.544% | 1.881 ± 0.173% |
| | 32 | 3.570 ± 1.263% | 2.404 ± 0.542% | 1.763 ± 0.161% | 1.501 ± 0.295% |
| | 64 | 3.902 ± 1.115% | 1.924 ± 0.110% | 1.557 ± 0.191% | 1.447 ± 0.194% |
| | 128 | 3.433 ± 0.363% | 1.697 ± 0.137% | 1.534 ± 0.151% | 1.556 ± 0.130% |
| | 256 | 3.415 ± 0.213% | 1.700 ± 0.134% | 1.540 ± 0.089% | 1.302 ± 0.283% |
| | 512 | 3.523 ± 0.316% | 1.551 ± 0.104% | 1.462 ± 0.279% | 1.379 ± 0.250% |
| | 1024 | 4.836 ± 1.532% | 3.537 ± 2.186% | 1.535 ± 0.128% | 1.404 ± 0.223% |
| 16 | 16 | 3.480 ± 0.856% | 3.136 ± 0.267% | 2.024 ± 0.194% | 1.734 ± 0.293% |
| | 32 | 3.636 ± 1.546% | 2.138 ± 0.362% | 1.605 ± 0.225% | 1.634 ± 0.258% |
| | 64 | 3.107 ± 0.079% | 1.696 ± 0.206% | 1.646 ± 0.128% | 1.363 ± 0.165% |
| | 128 | 3.229 ± 0.198% | 1.796 ± 0.204% | 1.612 ± 0.081% | 1.379 ± 0.245% |
| | 256 | 3.308 ± 0.266% | 1.701 ± 0.279% | 1.521 ± 0.083% | 1.322 ± 0.307% |
| | 512 | 3.474 ± 0.225% | 3.546 ± 2.577% | 1.556 ± 0.372% | 1.513 ± 0.234% |
| | 1024 | 3.337 ± 0.238% | 2.119 ± 0.798% | 1.946 ± 0.421% | 1.561 ± 0.056% |
| 32 | 32 | 3.017 ± 0.145% | 1.951 ± 0.447% | 1.574 ± 0.283% | 1.419 ± 0.265% |
| | 64 | 3.085 ± 0.306% | 1.672 ± 0.039% | 1.722 ± 0.168% | 1.331 ± 0.101% |
| | 128 | 3.207 ± 0.071% | 1.763 ± 0.169% | 1.433 ± 0.258% | 1.268 ± 0.149% |
| | 256 | 3.227 ± 0.114% | 1.704 ± 0.218% | 1.637 ± 0.226% | 1.457 ± 0.224% |
| | 512 | 3.335 ± 0.313% | 2.879 ± 0.667% | 2.079 ± 1.073% | 1.480 ± 0.285% |
| | 1024 | 3.075 ± 0.457% | 2.708 ± 0.700% | 1.819 ± 0.828% | 1.443 ± 0.439% |
| 64 | 64 | 2.769 ± 0.575% | 1.789 ± 0.086% | 1.640 ± 0.327% | 1.292 ± 0.226% |
| | 128 | 3.050 ± 0.381% | 1.654 ± 0.075% | 1.505 ± 0.225% | 1.402 ± 0.264% |
| | 256 | 3.028 ± 0.171% | 1.666 ± 0.158% | 1.398 ± 0.292% | 1.697 ± 0.295% |
| | 512 | 3.672 ± 1.109% | 2.351 ± 0.669% | 1.943 ± 0.668% | 1.649 ± 0.235% |
| | 1024 | 2.434 ± 0.591% | 2.019 ± 0.479% | 1.596 ± 0.206% | 1.475 ± 0.205% |
| 128 | 128 | 2.921 ± 0.339% | 1.597 ± 0.078% | 1.467 ± 0.375% | 1.354 ± 0.258% |
| | 256 | 3.039 ± 0.213% | 1.879 ± 0.711% | 1.701 ± 0.213% | 1.640 ± 0.284% |
| | 512 | 3.139 ± 0.132% | 1.578 ± 0.169% | 1.636 ± 0.417% | 1.681 ± 0.298% |
| | 1024 | 1.840 ± 0.126% | 1.618 ± 0.151% | 1.463 ± 0.175% | 1.692 ± 0.506% |
| 256 | 256 | 3.004 ± 0.093% | 1.894 ± 0.732% | 2.461 ± 1.016% | 2.333 ± 0.950% |
| | 512 | 2.684 ± 0.421% | 1.694 ± 0.207% | 1.679 ± 0.439% | 1.596 ± 0.285% |
| | 1024 | 1.841 ± 0.339% | 1.816 ± 0.292% | 1.602 ± 0.324% | 1.290 ± 0.323% |

The plot illustrates the generalization capability of the model on the Antiderivative operator benchmark. Data are presented as mean values ± SD derived from $n = 5$ independent trials with randomized network initializations and data splits.

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**Extended Data Table 4 | Ablation study on circuit architecture w.r.t. the Relative $L_2$ error**

| Qubit # | $h_t$ | 10 | 20 | 30 | 40 |
|---|---|---|---|---|---|
| | | Rel. $L_2$ Error | | | |
| 2 | 50 | 4.524 ± 1.073% | 3.805 ± 0.751% | 3.018 ± 0.160% | 3.105 ± 0.169% |
| | 100 | 3.280 ± 0.069% | 3.137 ± 0.252% | 2.842 ± 0.391% | 2.944 ± 0.401% |
| 5 | 20 | 2.414 ± 0.183% | 2.252 ± 0.119% | 2.212 ± 0.087% | 1.879 ± 0.144% |
| | 40 | 2.200 ± 0.087% | 1.761 ± 0.165% | 1.613 ± 0.135% | 1.638 ± 0.236% |
| 10 | 10 | 5.754 ± 0.263% | 3.744 ± 0.300% | 3.076 ± 0.096% | 2.597 ± 0.242% |
| | 20 | 3.413 ± 0.262% | 2.672 ± 0.217% | 2.306 ± 0.225% | 2.045 ± 0.113% |

Data are presented as mean values ± SD derived from $n$ = 5 independent trials with randomized network initializations and data splits.

 https://doi.org/10.1038/s42256-026-01289-7

**Extended Data Table 5 | Performance comparison between noise-free simulator and real-device (`ibm_fez`), alongside detailed circuit compilation statistics**

| $u(x)$ | Metric | Simulator | Real-device |
|---|---|---|---|
| $x$ | MSE | $1.534 \times 10^{-4}$ | $8.222 \times 10^{-3}$ |
| | Rel. $L_2$ Error | 5.498% | 40.247% |
| $\cos \pi x$ | MSE | $7.400 \times 10^{-5}$ | $5.190 \times 10^{-3}$ |
| | Rel. $L_2$ Error | 3.841% | 32.168% |
| Gate Complexity | Metric | Logical | Physical |
| Compilation | Circuit Depth | 60 | 109 |
| | Gate Breakdown | $R_y(40)$, $R_x(20)$ | $R_z(92)$, $\sqrt{X}(82)$ |
| | | $R_z(20)$, $CX(20)$ | $CZ(20)$ |
| | Total Count | 100 | 194 |

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https://doi.org/10.1038/s42256-026-01289-7

**Extended Data Table 6 | Default parameters for each problem. *P* denotes the number of sampling locations**

| Case | $u$ space | Train $u$ # | Train $P$ | Test $u$ # | Test $P$ | Batch Size | Iter. # |
|---|---|---|---|---|---|---|---|
| ODE | GRF($l$ = 0.2) | 1000 | 10 | 1000 | 100 | 100 | 100K |
| PDE | GRF($l$ = 0.2) | 1000 | 100 | 1000 | 1000 | 100 | 100K |

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https://doi.org/10.1038/s42256-026-01289-7

**Extended Data Table 7 | Hyperparameters for each network in the ODE case**

| Method | Branch | | Trunk | | Gates # | Param.# |
|---|---|---|---|---|---|---|
| | Depth $h_b$ | Ansatz Depth $l_b$ | Depth $h_t$ | Ansatz Depth $l_t$ | | |
| QuanONet | 20 | 2 | 20 | 2 | 1800 | 1201 |
| TF-QuanONet | 20 | 2 | 10 | 2 | 1350 | 1201 |
| | Depth | | Ansatz Depth | | | |
| HEA | 40 | | 2 | | 1800 | 1200 |
| TF-HEA | 32 | | 2 | | 1440 | 1280 |
| | Branch | | Trunk | | | |
| | Depth | Width | Depth | Width | | |
| DeepONet | 2 | 10 | 2 | 10 | — | 1261 |
| | Depth | | Width | | | |
| FNN | 2 | | 10 | | — | 1251 |

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 https://doi.org/10.1038/s42256-026-01289-7

**Extended Data Table 8 | Hyperparameters for each network in PDE case**

| Method | Branch | | Trunk | | Gates # | Param.# |
|---|---|---|---|---|---|---|
| | Depth $h_b$ | Ansatz Depth $l_b$ | Depth $h_t$ | Ansatz Depth $l_t$ | | |
| QuanONet | 40 | 2 | 40 | 2 | 3600 | 2401 |
| TF-QuanONet | 40 | 2 | 20 | 2 | 2700 | 2401 |
| | Depth | | Ansatz Depth | | | |
| HEA | 80 | | 2 | | 3600 | 2400 |
| TF-HEA | 64 | | 2 | | 2880 | 2560 |
| | Branch | | Trunk | | | |
| | Depth | Width | Depth | Width | | |
| DeepONet | 3 | 15 | 3 | 15 | — | 2521 |
| | Depth | | Width | | | |
| FNN | 3 | | 16 | | — | 2481 |

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https://doi.org/10.1038/s42256-026-01289-7

**Extended Data Table 9 | Hyperparameters for FNO comparison**

| Method | Branch | | Trunk | | Gates # | Param.# |
|---|---|---|---|---|---|---|
| | Depth $h_b$ | Ansatz Depth $l_b$ | Depth $h_t$ | Ansatz Depth $l_t$ | | |
| TF-QuanONet | 160 | 2 | 90 | 2 | 11250 | 10001 |
| | Branch | | Trunk | | | |
| | Depth | Width | Depth | Width | | |
| DeepONet | 4 | 32 | 4 | 32 | — | 9633 |
| | Modes # | Width | Layers # | fc_hidden | | |
| FNO | 19 | 9 | 3 | 43 | — | 10005 |

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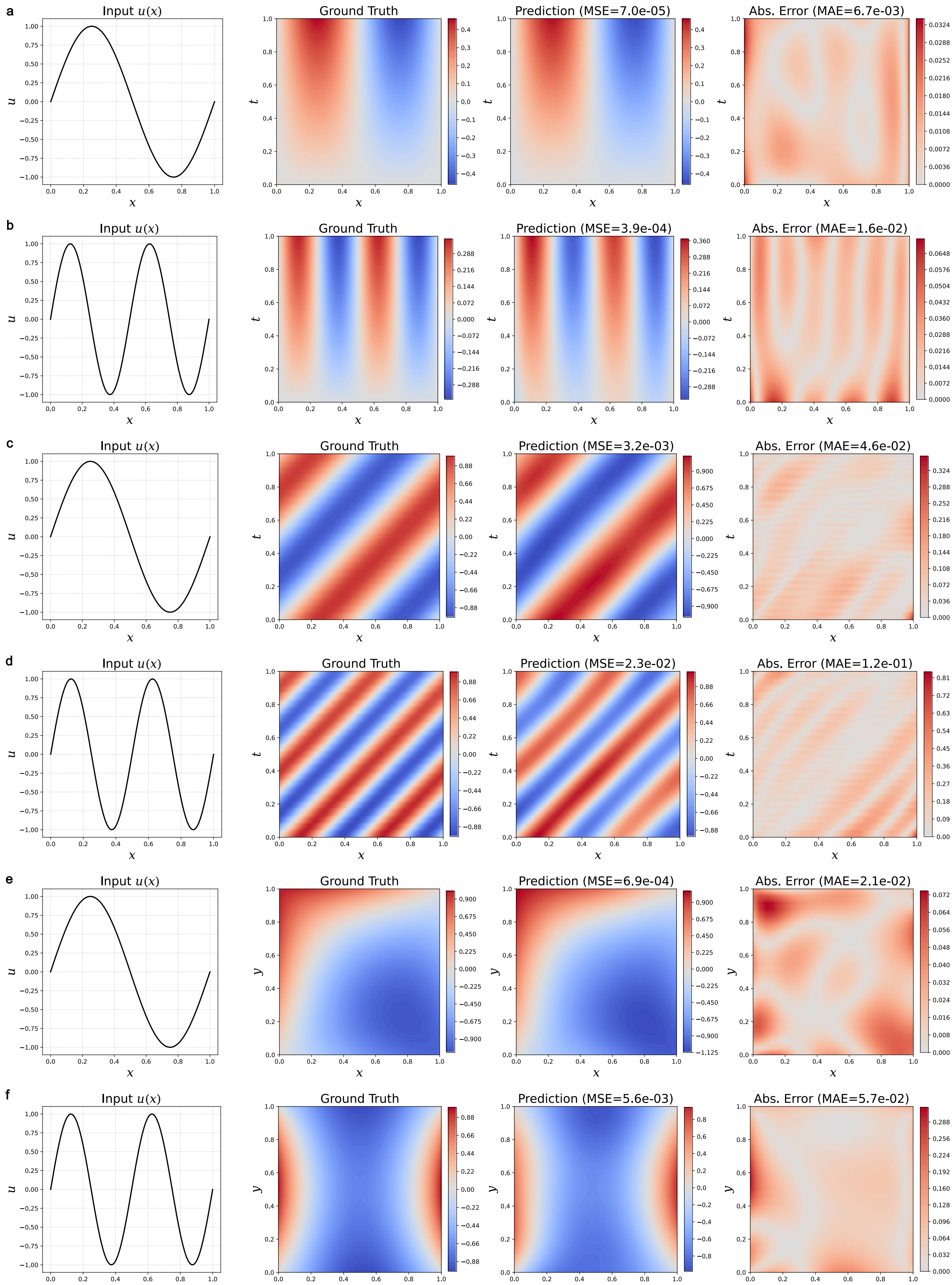
https://doi.org/10.1038/s42256-026-01289-7




**Extended Data Fig. 1 | Visualization of TF-QuanONet predictions versus ground truth.** The rows display results for the D-R System (**a**, **b**), Advection Equation (**c**, **d**), and Darcy Flow (**e**, **f**), corresponding to distinct input frequencies: $u(x) = \sin 2\pi x$ and $u(x) = \sin 4\pi x$.

nature portfolio

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Last updated by author(s): Jul 3, 2026

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*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | Synthetic datasets (Gaussian random fields) were generated using standard numerical solvers: the Runge-Kutta (4,5) method for ODE systems and a second-order finite difference scheme for PDEs, implemented in Python using standard libraries (e.g., NumPy, SciPy). |
| Data analysis | Quantum neural network simulations were performed using the MindSpore framework and MindQuantum. Classical neural network baselines were implemented using the DeepXDE library based on PyTorch. Real quantum hardware experiments were conducted on the IBM Quantum processor (ibm_fez) using the Qiskit SDK. |

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The datasets generated and analysed during the current study are purely synthetic, derived directly from the code using standard numerical solvers. No proprietary, clinical, or third-party datasets were used. The data generation scripts, along with the pre-generated training and testing datasets necessary to reproduce the results, are publicly available in the GitHub repository at https://github.com/Wang-Ruocheng/QuanONet and have been permanently archived on Zenodo at https://doi.org/10.5281/zenodo.21084057.

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| | |
|---|---|
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|---|---|
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| | |
|---|---|
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| | |
|---|---|
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| Sampling strategy | *Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.* |
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| ☒ | ☐ Animals and other organisms |
| ☒ | ☐ Clinical data |
| ☒ | ☐ Dual use research of concern |
| ☒ | ☐ Plants |

## Methods

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| | |
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